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A. A Case Study of Disagreement Between Papers

The wide variety of non-equivalent definitions for model collapse often creates apparent contradictions between papers claiming to study the same phenomenon. A representative example presents itself in the study of model collapse for linear regression. In this data setting, one begins with a dataset (X, y) where we assume that

$$y \sim X\beta + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma \cdot I).$$

In the first iteration, one computes

$$\hat{\beta}^{(1)} = (X^T X)^{-1} X^T y,$$

and uses the fit parameter to generate new data $(X, y^{(1)})$ with

$$y^{(1)} = X\hat{\beta}^{(1)} + \epsilon_1, \epsilon_1 \sim \mathcal{N}(0, \sigma \cdot I).$$

One can then fit successive model iterations using an *accumulate* paradigm, in which the data for the n th model fitting takes the form

$$\left(\begin{bmatrix} y, y^{(1)}, y^{(2)}, \dots, y^{(n-1)} \end{bmatrix}^T, \begin{bmatrix} X, X, \dots, X \end{bmatrix}^T \right)$$

One can also fit the n th model iteration using a *replace* paradigm, in which $\hat{\beta}^{(n)}$ is computed using only the data $(X, y^{(n-1)})$. As proven by Gerstgrasser et al. (2024), in the accumulate paradigm, the ratio

$$\frac{\mathbb{E} \left[\|\hat{\beta}^{(n)} X_{\text{test}} - y_{\text{test}}\|^2 \right]}{\mathbb{E} \left[\|\hat{\beta}^{(1)} X_{\text{test}} - y_{\text{test}}\|^2 \right]}$$

monotonically increases before converging to $\pi^2/6$. Citing Definitions 3 and 4, Gerstgrasser et al. (2024) correctly asserted that model collapse does not occur. However, if one instead defines model collapse by Definition 2, this scenario does exhibit collapse.

To complicate the story, Dohmatob et al. (2024a) studied the same model under the replace paradigm. Under the replace paradigm, Definition 3 suggests that model collapse occurs since the asymptotic risk diverges, while Definition 4 implies that model collapse does not occur, since the replace scenario does not exhibit vanishing variance. Table 1 shows how incompatible definitions lead to confusion.

Table 1. Model collapse occurs under some definitions, but does not occur under others for the regression setting described in Appendix A.

Definition	Accumulate	Replace
Def. 1	✗	✓
Def. 2	✓	✓
Def. 3	✗	✓
Def. 4	✗	✗
Def. 5	✓	✓
Def. 6	✗	✗
Def. 7	✗	✗
Def. 8	✗	✗